

ORDER-MINIMAL PENDANT-FREE COUNTEREXAMPLES TO CONJECTURE 2 OF NURDIN ET AL.

ABSTRACT. Let $B(G)$ be the degree-count lower bound conjectured by Nurdin et al. to be equal to the total vertex irregularity strength $\text{tvs}(G)$. We determine all connected simple graphs with minimum degree at least two and order at most seven for which $\text{tvs}(G) > B(G)$. Exactly three occur, all on seven vertices: $K_{3,4}$ and two explicitly given non-complete-multipartite graphs. Each has $B(G) = 2$ and $\text{tvs}(G) = 3$. The finite classification is certified by explicit $B(G)$ -labelings for the other 580 atlas graphs, while the three obstructions have elementary proofs. We also give a regular-core join construction which, together with the seven-vertex examples, yields non-complete-multipartite counterexamples of every order at least seven. Since $K_{3,4}$ appears to be covered already by a 2005 complete-bipartite result, the contribution is the order-minimal classification and the regular-core extension, not the first counterexample.

1. RESULTS

Let $G = (V, E)$ be a finite simple connected graph. A *vertex-irregular total k -labeling* is a map

$$\phi : V \cup E \longrightarrow \{1, \dots, k\}$$

for which the vertex weights

$$\text{wt}_\phi(v) = \phi(v) + \sum_{uv \in E} \phi(uv)$$

are pairwise distinct. The least such k is the *total vertex irregularity strength* $\text{tvs}(G)$, introduced by Bača et al. [1].

For each integer j , let n_j be the number of vertices of degree j , and let δ and Δ be the minimum and maximum degrees. Nurdin et al. [2] proved the lower bound

$$B(G) := \max_{\delta \leq i \leq \Delta} \left\lceil \frac{\delta + \sum_{j=\delta}^i n_j}{i+1} \right\rceil \leq \text{tvs}(G)$$

and conjectured equality in [2, Conjecture 2]. Susanto, Simanjuntak, and Baskoro later produced infinite counterexample families and asked whether a counterexample can have no pendant vertices [4, Question 4.2]. That existence question may already have been settled by the earlier complete-bipartite work of Wijaya et al. [5], which appears to contain $\text{tvs}(K_{3,4}) = 3$,

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whereas $B(K_{3,4}) = 2$; we have not been able to consult that paper, and Susanto et al. do not cite it when posing Question 4.2. We therefore make no priority claim for pendant-free existence.

Our first result gives the exact minimum-order classification.

Theorem 1. *Among connected simple graphs G with $\delta(G) \geq 2$ and $|V(G)| \leq 7$, the inequality $\text{tvs}(G) > B(G)$ holds for exactly three isomorphism classes. They are $K_{3,4}$ and the graphs G_1, G_2 on $\{0, \dots, 6\}$ with*

$$E(G_1) = \{01, 12, 23, 04, 34, 15, 35, 06, 16\},$$

$$E(G_2) = \{01, 02, 23, 14, 34, 05, 45, 06, 36\},$$

where ij denotes the edge $\{i, j\}$. For all three graphs,

$$B(G) = 2 \quad \text{and} \quad \text{tvs}(G) = 3.$$

The graph6 strings of G_1, G_2 , and the atlas representative of $K_{3,4}$ are, respectively,

$$\text{FhdU?}, \quad \text{FpUK}_-, \quad \text{FreRW}.$$

The graphs G_1 and G_2 are nonisomorphic: G_1 contains a triangle, whereas G_2 is triangle-free. Neither is complete multipartite, since in each graph $0 \not\sim 3$ and $3 \not\sim 1$, but $0 \sim 1$; nonadjacency is transitive in every complete multipartite graph.

Our second result extends the endpoint-weight obstruction from complete multipartite graphs to an arbitrary regular core.

Theorem 2. *Let $t \geq 1$ and $d \geq \max\{2t, t+2\}$. If H is a $(d-t-2)$ -regular simple graph on d vertices, then*

$$G = H \vee \overline{K}_{2t+2}$$

satisfies $B(G) = 2 < \text{tvs}(G)$.

Corollary 3. *For every $n \geq 7$, there is a connected non-complete-multipartite graph G of order n and minimum degree at least two such that $B(G) = 2 < \text{tvs}(G)$.*

2. THE THREE ORDER-MINIMAL GRAPHS

The degree sequence of each of G_1 and G_2 is $(2, 2, 2, 2, 3, 3, 4)$. Hence

$$B(G_i) = \max \left\{ \left\lceil \frac{2+4}{3} \right\rceil, \left\lceil \frac{2+6}{4} \right\rceil, \left\lceil \frac{2+7}{5} \right\rceil \right\} = 2.$$

Proposition 4. *Neither G_1 nor G_2 admits a vertex-irregular total 2-labeling.*

Proof. Suppose that G_i has such a labeling. Its four degree-2 vertices have four distinct weights in $\{3, 4, 5, 6\}$, so they use all four values. The two degree-3 vertices must then have weights 7 and 8, and the unique degree-4 vertex c has weight 9 or 10.

Let z be the degree-2 vertex of weight 3. The label of z and both incident edge labels are 1. In G_2 , every degree-2 vertex is adjacent to c and to one

degree-3 vertex. In G_1 , exactly one degree-2 vertex is adjacent to both degree-3 vertices; it cannot be z , because the degree-3 vertex of weight 8 and all its incident edges have label 2. Thus, in either graph, z is adjacent to c and to a degree-3 vertex a .

The edge cz has label 1, so c cannot have weight 10. Hence $\text{wt}(c) = 9$, and every contribution at c other than cz is 2. Similarly, $\text{wt}(a) = 7$, and every contribution at a other than az is 2. The other degree-3 vertex has weight 8, so all four of its contributions are 2. From the displayed edge sets, every degree-2 vertex other than z has both incident edges among these forced label-2 edges. Its weight is therefore at least $1 + 2 + 2 = 5$, contradicting the required degree-2 weight 4. $\square$

For $K_{3,4}$,

$$B(K_{3,4}) = \max \left\{ \left\lceil \frac{3+4}{4} \right\rceil, \left\lceil \frac{3+7}{5} \right\rceil \right\} = 2.$$

A total 2-labeling would give seven distinct weights in the seven-integer interval $[4, 10]$, so the weights would exhaust that interval. Weight 4 forces a degree-3 vertex and its three incident edges to have label 1, while weight 10 forces a degree-4 vertex and its four incident edges to have label 2. Those two vertices are adjacent, a contradiction.

For completeness, the following certificates give total 3-labelings. Vertices are ordered $0, 1, \dots, 6$; edge labels are read in the displayed edge order for G_1, G_2 , and in the order

03, 04, 05, 06, 13, 14, 15, 16, 23, 24, 25, 26

for $K_{3,4}$ with bipartition $\{0, 1, 2\} \cup \{3, 4, 5, 6\}$.

Graph	vertex labels	edge labels	vertex weights
G_1	3313233	111111123	7, 9, 3, 6, 4, 5, 8
G_2	3123333	111111132	9, 3, 4, 7, 6, 5, 8
$K_{3,4}$	2221222	111211231233	7, 9, 11, 4, 6, 8, 10

Thus all three graphs have total vertex irregularity strength 3.

Proof of Theorem 1. The Read–Wilson graph atlas [3] contains one representative of every simple graph of order at most seven. Filtering for connected graphs of minimum degree at least two gives

$ V(G) $	3	4	5	6	7
graphs	1	3	11	61	507.

Each of the other 580 graphs admits an explicit vertex-irregular total $B(G)$ -labeling, located by exhaustive search as follows. Writing $\text{wt}_\phi(v) = 1 + \deg(v) + a_v + b_v$ with $a_v = \phi(v) - 1 \in [0, k - 1]$ and $b_v = \sum_{e \ni v} (\phi(e) - 1)$, one enumerates the $k^{|E|}$ vectors of edge offsets $\phi(e) - 1$, which determine the b_v , and tests by greedy interval matching on the length- k intervals $[\deg(v) + b_v, \deg(v) + b_v + k - 1]$ whether the a_v can be chosen to make the weights pairwise distinct. Run over all 583 graphs of the census at $k = B(G)$, this

search returns a labeling for every graph except **FhdU?**, **FpUK₋**, and **FreRW**, and terminates in well under a second; it needs no ancillary data. The preceding hand proofs establish infeasibility at $k = 2$ for those three graphs. Hence the classification is complete. $\square$

3. A REGULAR-CORE CONSTRUCTION

Proof of Theorem 2. The $2t + 2$ vertices in the independent set have degree d , while each vertex of H has degree

$$(d - t - 2) + (2t + 2) = d + t.$$

Thus G has order $n = d + 2t + 2$, minimum degree d , and maximum degree $d + t$. For $d \leq i < d + t$, the cumulative degree count is $2t + 2$, and

$$1 < \frac{d + 2t + 2}{i + 1} \leq \frac{d + 2t + 2}{d + 1} \leq 2.$$

At $i = d + t$, all vertices have been counted and

$$\frac{d + n}{d + t + 1} = \frac{2d + 2t + 2}{d + t + 1} = 2.$$

Therefore $B(G) = 2$.

Assume that a vertex-irregular total 2-labeling exists. The two degree classes have weight intervals

$$[d + 1, 2d + 2] \quad \text{and} \quad [d + t + 1, 2d + 2t + 2].$$

These intervals overlap, since $d + t + 1 \leq 2d + 2$, and their union is the interval $[d + 1, 2d + 2t + 2]$. It contains exactly

$$(2d + 2t + 2) - (d + 1) + 1 = d + 2t + 2 = n$$

integers. The n distinct vertex weights must therefore exhaust this interval. Its minimum forces a degree- d vertex and all its incident edges to have label 1; its maximum forces a degree- $(d + t)$ vertex and all its incident edges to have label 2. These vertices lie on opposite sides of the join and are adjacent, so their common edge would have both labels. This contradiction proves $\text{tvs}(G) > 2$. $\square$

Proof of Corollary 3. For $n = 7$, take G_1 . For $n \geq 8$, set $t = 1$, $d = n - 4$, and $H = \overline{C_d}$. Then H is $(d - 3)$ -regular, so Theorem 2 applies. Moreover,

$$\overline{H \vee \overline{K_4}} = C_d \cup K_4.$$

Since $d \geq 4$, this complement is not a disjoint union of cliques; hence the resulting graph is not complete multipartite. $\square$

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